

- 2 A program sorts the data in the 1D array `DataArray` and searches `DataArray` for specific values.

`DataArray` stores 14 integer values and is declared local to the main program.

- (a) Write program code to create `DataArray` and initialise it with the following data values in the order they are written:

0 3 4 56 67 44 43 32 31 345 45 6 54 1

Save your program as **Question2_J25**.

Copy and paste the program code into part **2(a)** in the evidence document.

[1]

- (b) The function `InsertionSort()` takes an array of integers as a parameter. The function sorts the data in the array into ascending numerical order using an insertion sort. The function returns the sorted array.

Write program code for `InsertionSort()`

You must **not** use any inbuilt sorting functions for your programming language.

Save your program.

Copy and paste the program code into part **2(b)** in the evidence document.

[5]

- (c) The procedure `OutputArray()` takes an array of integers as a parameter. The procedure outputs each element in the array from the first element to the last element. The output is on one line with a space between each number.

An example output is:

"0 3 4 56 67 44 43 32 31 345 45 6 54 1"

Write program code for `OutputArray()`

Save your program.

Copy and paste the program code into part **2(c)** in the evidence document.

[2]